

DETAILED DESCRIPTION OF THE INVENTION — MANUFACTURING METHODS, MATERIAL SELECTION, AND SCALABLE PRODUCTION VARIANTS

The intraluminal physiologic monitoring device may be manufactured using a variety of production techniques and material systems selected to achieve comfort, durability, biocompatibility, and scalable manufacturing efficiency. The invention therefore encompasses multiple fabrication approaches and production variants.

In various embodiments, device components may be fabricated using injection molding, overmolding, compression molding, additive manufacturing, extrusion processes, casting methods, or hybrid fabrication techniques combining rigid and compliant structures.

Material selections may include medical-grade silicone, thermoplastic elastomers, polyurethane materials, biocompatible polymers, flexible composites, metallic alloys, textile-reinforced elastomers, or multilayer laminated constructions designed to optimize flexibility, strength, and tactile comfort.

Certain embodiments may incorporate soft-touch exterior layers surrounding rigid internal frames protecting electronic components while maintaining user comfort. Multi-durometer constructions may combine regions of differing stiffness within a single device body.

Manufacturing methods may include encapsulation of sensors within sealed elastomeric layers, conformal coating of electronic assemblies, ultrasonic welding, adhesive bonding, laser sealing, or thermal fusion processes configured to achieve waterproof and hygienic device construction.

Scalable production variants may allow simplified consumer models utilizing reduced sensor sets alongside advanced clinical models incorporating expanded sensing capability. Production flexibility may support multiple price tiers without departing from the core invention.

The pull tab assembly and externalized electronics structures may be manufactured separately and joined during final assembly using modular connectors, magnetic couplings, or integrated flexible interconnects.

Disposable or semi-disposable components may be manufactured using cost-efficient processes enabling hygienic replacement while reusable electronics remain protected within sealed housings.

Future manufacturing approaches may incorporate automated robotic assembly, microfabrication techniques, printed electronics, stretchable circuits, or emerging material technologies compatible with evolving device architectures.

The manufacturing methods and material selections described herein are illustrative rather than limiting. Any fabrication process or material configuration capable of producing an intraluminal physiologic monitoring device is considered within the scope of the present disclosure.